

UNIVERSITY OF CALCUTTA
Mode of Examination: Online
M.Sc. (COMPUTER SCIENCE) Semester- I Examination, 2021

Subject: Computer Science
Paper Code & Name: CSMC103 & ADVANCED COMPUTER ARCHITECTURE

Date: 07.03.2022

Time and Duration: 4:30PM-7:30PM (3:00 Hours)

Please note the following instructions carefully:

- **Promise not to commit any academic dishonesty.**
 - **Marks will be deducted if the same/similar answers are found in different answer scripts.**
 - **Candidates are required to answer in their own words as far as applicable.**
 - **The figures in the margin indicate full marks.**
 - **Each page of the answer-scripts should have your University Roll # on the right-top corner.**
 - **The name of the scanned copy of the answer script will be of the following format:**
CSMC103-ACA- University-Roll-Number.pdf
(Example:-CSMC103-ACA-C91-CSC-2110xy.pdf)
 - **The subject of the mail should be the file name only.**
 - **The scanned answer script is to be sent to cucse2020@gmail.com**
 - **The file should have the top page (Page #1) as an index page, mentioning all page numbers against the answer of each question number.**
 - **The answer script may not be accepted after the scheduled time.**
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Answer **Question Nos. 1 and 2**, and **any four** from the rest.

- 1. Answer any five questions:** **[2x5]**
- i. Which deficiencies of CISC justify the development of RISC?
 - ii. What do you mean by Instruction Set Architecture? What are its implications?
 - iii. What are vertical and horizontal microprogramming?
 - iv. Explain why cache hit ratios are not an objective measurement of cache effectiveness.
 - v. Comment on Clock skewing.
 - vi. What are the different types of internal data forwarding? Give examples.
 - vii. How many states are there in 4x4 switch, including broadcasting and permutation? Comment on the control line format.
- 2. Answer any five questions:** **[4x5]**
- i. What reasons for the introduction of delay stages in nonlinear pipeline architecture may reduce the synthesized minimal average latency? Justify your answer.
 - ii. Briefly comment on the major reasons for the cache coherence problem in any multiprocessor system.
 - iii. Estimate the effect of branch instructions on the performance of pipelined architectures with a suitable set of parameters.
 - iv. 'Multilevel cache memory implementation is an example of super-linearity' - critically comment on the statement.
 - v. Split out the design steps to get a Reservation Table (RT) for a given latency cycle (not constant). Explore the design steps for the latency cycle (2,3,2,5).

- vi. Write an algorithm to add 20 numerical values using a 3-cube static processor setup. State your assumptions.
- vii. A computer has a 16-way interleaved memory. We are required to access a 64 by 64 matrix. Compute the total time needed for the access, if the elements are accessed by
 - a. row by row,
 - b. column by column.

- 3. a. What is the significance of virtual memory in advanced computer architecture? Justify.
- b. What is memory interleaving? Briefly explain two interleaved memory organizations with suitable assumptions.
- c. The memory unit has 256 K words of 32 bits each. The computer has an instruction format with four fields, an op-code field, a mode to specify one of six modes, a register address field to specify one of 56 processor registers, and a memory address. Specify the instruction format. How many instructions are there at most on the computer?
- d. Assume cache size is 64 KB, block size is 32 B, and the cache is Two-way Set associative. For a 32-bit physical address, give the division between Block offset, Index, and Tag bits.

[2+3+3+2]

- 4. a. Let, A be a $2^k \times 2^k$ matrix stored in row-major order in the main memory. Prove that the transposed matrix A^T can be obtained by performing k perfect shuffles on A.
- b. Identify dependencies among the following statements in a given program:

S1:	LOAD R1, 20H	/ R1 < 20H /
S2:	LOAD R2, M (10H)	/ R2 < MEMORY (10H) /
S3:	ADD R1, R2	/ R2 < (R1) +(R2) /
S4:	STORE M (20H), R1	/ MEMORY (20H) < R1 /
S5:	STROE M ((R2)), 20H	/ MEMORY ((R2)) < 20H /

Explain your answer. Draw the dependency graph.
- c. Critically comment on the states of a process of any message passing system.

[4+4+2]

- 5. a. "Algorithm depends on the underlying architecture" – Critically comment on the statement with a suitable example.
- b. Is it possible for the average speedup exhibited by a parallel algorithm to be superlinear? Justify your answer.
- c. Prove that the number of necklaces in an k-bit shuffle exchange network is $O(2^k/k)$.
[The nodes through which a data item travels in response to a sequence of shuffles are called necklace of i, where i is the start-address]

[3+3+4]

- 6. a. Write an algorithm to identify the conflicting pairs of source-destination for a shuffle exchange MICN of a given size.
- b. Prove that a MICN with $\log_2 N$ stages and each stage with 2×2 ($N/2$) switches is a blocking interconnection network. What are the alternative ways to transform it to a non-blocking MICN?
- c. Comment on the permutation mapping capability of Multi-stage Interconnection Networks.

[4+4+2]

- 7. a. For the given reservation table of a pipeline architecture, what are the theoretical lower limit and

the theoretical upper limit of MAL values? Justify your answer.

- b. Why does the synthesized value of MAL for a given reservation table differs from the theoretical lower limit value of MAL? Justify your answer.
 - c. What are the different pipeline hazards? How do you avoid them? **[5+2+3]**
- 8.
- a. Compare and contrast 'Systolic Architecture' with SIMD architecture.
 - b. What are the desired functional properties of any systolic architecture? Briefly justify your claims.
 - c. What down the standard algorithm for matrix multiplication on a uni-processor system and modify the algorithm for systolic architecture with N^2 PEs, where the size of the matrices are $N \times N$. **[2+2+6]**